

P3016R2

Resolve inconsistencies in `begin/end` for `valarray` and braced initializer lists

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Abstract

We resolve some inconsistencies among `[iterator.range]` functions, notably `begin/end`, `data`, and `empty`, as applied to `valarray`, `initializer_list`, and braced initializer lists. We also resolve LWG3624 and LWG3625.

Table of Contents

1	Changelog
2	Motivation and proposal
2.1	<code>data</code> and <code>empty</code>
2.2	Availability of <code>begin/end</code>
3	Implementation experience
3.1	Tony Table
4	Proposed wording
4.1	[version.syn]
4.2	[valarray.syn]
4.3	[template.valarray.overview]
4.4	[valarray.members]
4.5	[support.initlist]
4.6	[iterator.synopsis]
4.7	[iterator.range]
4.8	<code>.data+.size</code> cleanup
4.8.1	[string.cons]
4.8.2	[string.append]
4.8.3	[string.assign]

- 4.8.4 [string.replace]
- 4.8.5 [span.cons]
- 4.8.6 [valarray.cons]
- 4.9 Resolutions for LWG3624 and LWG3625

References

Informative References

§ 1. Changelog

- R2:
 - Give [resolutions for LWG3624 and LWG3625](#)
 - Rebut concerns about [breaking code](#); discuss how a deprecation period for `std::begin(initializer_list)` is basically unimplementable
 - Make `valarray::iterator` non-exposition-only
 - Remove R1's [Historical Background](#) section, as basically distracting at this point
 - Mention [\[P2613\]](#) and [\[LWG4035\]](#) as further precedent for `initializer_list.empty()`
 - Add feature-test macros `__cpp_lib_initializer_list` and `__cpp_lib_valarray`
- R1:
 - Split [\[CWG2825\]](#) from this paper; this paper is now library-only
 - Add discussion of `.data()` and `.empty()` overloads
 - Add wording for ".data+.size cleanup"

§ 2. Motivation and proposal

Casey Carter [points out](#) that the following program is supported by libstdc++ but not libc++ nor Microsoft ([Godbolt](#)):

```
#include <iterator>
#include <valarray>
int main() {
    std::valarray<int> v = {1,2,3};
    std::begin(v); // OK
    std::cbegin(v); // Error
}
```

This is because `std::valarray` defines its own non-member, non-hidden-friend overloads of `std::begin` and `std::end`. These overloads are found by the qualified call to `std::begin` here, but aren't found by `std::cbegin`'s ADL because the primary template for `std::cbegin` happens to be defined before `<valarray>` is included. Swapping the order of `#include <iterator>` and `#include <valarray>` in this example doesn't help, because the relevant parts of `<iterator>` are still transitively included by `<valarray>` before `std::valarray`'s own code.

Likewise, on all vendors ([Godbolt](#)):

```
#include <iterator>
int main() {
    std::begin({1,2,3}); // OK
    std::cbegin({1,2,3}); // Error
}
```

This is because `{1,2,3}` is a *braced-initializer-list* with no type; so it cannot bind to the deduced `const C&` in `std::cbegin` (defined in `<iterator>`). But it *can* bind to the `initializer_list<E>` in the non-member, non-hidden-friend overload `std::begin(std::initializer_list<E>)` (defined in `<initializer_list>`).

Notice that `std::begin({1,2,3})` returns an iterator that will dangle at the end of the full-expression, and that the return values of `std::begin({1,2,3})` and `std::end({1,2,3})` do not form a range (because the two lists' backing arrays may be different). Therefore this overload is more harmful than helpful.

NOTE: Be careful to distinguish the scenario of calling `std::begin` on an *object of type `initializer_list`* (helpful!) from calling it on a *braced-initializer-list* (harmful).

We propose to resolve `valarray`'s `begin/cbegin` inconsistency in favor of "make it work" and to resolve *braced-initializer-list*'s `begin/cbegin` inconsistency in favor of "make it ill-formed."

§ 2.1. data and empty

We also propose two more member functions for `std::initializer_list`: `.data()` and `.empty()`.

Many places in the library clauses would like to operate on the contiguous data of an `initializer_list` using the "data + size" idiom, but since `initializer_list` lacks `.data()`, they're forced to use an awkward "begin + size" approach instead. This (1) looks unnatural and (2) causes extra mental effort for library writers. Part of P3016's proposed wording is to update these places in the library. For example:

```
constexpr basic_string& append(const basic_string& str);
1. Effects: Equivalent to return append(str.data(), str.size());
[...]

constexpr basic_string& append(initializer_list<charT> il);
16. Effects: Equivalent to return append(il.data(), il.size() il.begin(), il.size());
```

As for `.empty()`, it is generally recognized these days that ranges providing `.size()` should also provide `.empty()`. For example, `iota_view`'s missing `.empty()` was added as a DR by [\[LWG4001\]](#). [\[LWG4035\]](#) adds `single_view`'s missing `.empty()`. [\[P2613\]](#), which added `mdspan`'s missing `.empty()`, claims that `initializer_list` and `bitset` remain the only two types with `.size()` and without `.empty()`.

By making `il.data()` and `il.empty()` well-formed for `std::initializer_list` objects, we satisfy the SFINAE conditions of the primary templates for `std::data(il)` and `std::empty(il)`, meaning that we can eliminate their special overloads for `initializer_list` arguments.

<pre>std::initializer_list<int> il = {1,2,3}; const int *p = std::data(il); // OK, calls the initializer_list overload // specially provided in <iterator> bool b = std::empty(il); // OK, calls the initializer_list overload // specially provided in <iterator></pre>	<pre>std::initializer_list<int> il = {1,2,3}; const int *p = std::data(il); // OK, calls the primary template bool b = std::empty(il); // OK, calls the primary template</pre>
<pre>const int *p = std::data({1,2,3}); // calls the initializer_list overload // specially provided in <iterator>, // and immediately dangles const int *q = std::data({}); // does not compile</pre>	<pre>const int *p = std::data({1,2,3}); // does not compile const int *q = std::data({}); // does not compile</pre>
<pre>bool b = std::empty({1,2,3}); // calls the initializer_list overload // specially provided in <iterator>, // invariably returns false bool c = std::empty({}); // does not compile</pre>	<pre>bool b = std::empty({1,2,3}); // does not compile bool c = std::empty({}); // does not compile</pre>

§ 2.2. Availability of `begin/end`

The following snippet tries to use `std::begin(il)` without first including `<iterator>` nor any of the 15 other headers listed in [\[iterator.range\]/1](#) that declare the generic `std::begin`. This code works in C++23.

We propose simply to break the left-hand snippet, and force the programmer to write the right-hand snippet instead. One LEWG reviewer expressed concern that we were breaking valid code without a deprecation period.

<pre>#include <initializer_list> std::initializer_list<int> il = {1,2,3}; const int *b = std::begin(il); // OK, calls the initializer_list overload // specially provided in <initializer_list> const auto r = std::rbegin(il); // Error, not declared in <initializer_list> const int *c = std::cbegin(il); // Error, not declared in <initializer_list> size_t s = std::size(il); // Error, not declared in <initializer_list></pre>	<pre>#include <iterator> std::initializer_list<int> il = {1,2,3}; const int *b = std::begin(il); // OK, calls the primary template const auto r = std::rbegin(il); // OK, calls the initializer_list overload // specially provided in <iterator> const int *c = std::cbegin(il); // OK, calls the primary template size_t s = std::size(il); // OK, calls the primary template</pre>
--	---

At first, I was amenable to the idea that the declarations of `std::begin` and `std::end` in `<initializer_list>` should merely be deprecated, not fully removed in C++26, so that the behavior of the code on the left would be (temporarily) preserved. To do that, we'd modify `[iterator.range]` like this:

```
template<class C> constexpr auto begin(C& c) -> decltype(c.begin());
template<class C> constexpr auto begin(const C& c) -> decltype(c.begin());
```

2. Returns: `c.begin()`.

x. Remarks: In addition to being available via inclusion of the `<iterator>` header, these function templates are available when `<initializer_list>` is included. This availability is deprecated.

But library vendors can't implement that wording! We have `[[deprecated]]` to mark an *entity* as deprecated, but we have no way to mark a *single declaration* as deprecated. (I.e., "If, hypothetically, removing this declaration would make the call ill-formed, then give a deprecation warning; otherwise don't.") Three years from now, we'd come back asking "Can we remove this deprecated feature yet?" and the answer would be "No, vendors haven't started giving a warning for it yet," because it's physically impossible to give a warning for it. This would be a silly situation to get into. And the feature itself is so obscure (using `std::begin` specifically on an `initializer_list` without including any of more than a dozen STL headers) and so specific (as shown, it already doesn't work for `cbegin`, `rbegin`, or `size`), and the fix in code is so surgical (include `<iterator>` or `<span>`) that I don't think the lack of deprecation period will matter.

[LWG3624] and [LWG3625] somewhat intersect this proposal, but do not conflict with it. We expect these issues to be resolved along the lines given in § 4.9 Resolutions for LWG3624 and LWG3625.

§ 3. Implementation experience

Arthur has implemented § 4 Proposed wording in his fork of libc++ ([source](#)), and used it to compile both LLVM/Clang/libc++ and another large C++17 codebase. Naturally, it caused no problems except in this single test from libc++'s own test suite:

```
#include <initializer_list> // but not <iterator>
std::initializer_list<int> il;
static_assert(noexcept(std::begin(il)));
```

This test now fails first because `<iterator>` was not included, and second because today's `begin(initializer_list<E>)` is noexcept but the primary template `begin(C&)` is non-noexcept (per [P0884](#) guidance).

`il.begin()` and `ranges::begin(il)` remain noexcept.

`std::begin(valarray)` remains non-noexcept.

§ 3.1. Tony Table

<pre>#include <initializer_list> void f(std::initializer_list<int> il) { auto it = std::begin(il); }</pre>	<pre>#include <initializer_list> #include <iterator> // for std::begin void f(std::initializer_list<int> il) { auto it = std::begin(il); }</pre>
<pre>S(std::initializer_list<int> il) : S(il.begin(), il.size()) {}</pre>	<pre>S(std::initializer_list<int> il) : S(il.data(), il.size()) {}</pre>
<pre>auto dangle = std::begin({1,2,3}); // calls the initializer_list overload // specially provided in <initializer_list>, // and immediately dangles</pre>	<pre>// no longer compiles</pre>
<pre>bool b = std::empty({1,2,3}); // calls the initializer_list overload // specially provided in <iterator>, // yielding false</pre>	<pre>// no longer compiles</pre>
<pre>#include <valarray> #include <utility> std::valarray<int> va; auto it = std::begin(std::as_const(va));</pre>	<pre>#include <valarray> #include <iterator> std::valarray<int> va; auto it = std::cbegin(va);</pre>

§ 4. Proposed wording

NOTE: Vendors should provide `__cpp_lib_initializer_list` if `il.data()` and `il.empty()` are well-formed. Vendors should provide `__cpp_lib_valarray` if `va.begin()` and `va.end()` are well-formed and the free-function `begin` and `end` have been removed from `<valarray>`.

§ 4.1. [version.syn]

Add two feature-test macros to [version.syn]/2:

```
#define __cpp_lib_incomplete_container_elements 201505L // also in <forward_list>, <list>, <vector>
#define __cpp_lib_initializer_list YYYYMMML // also in <initializer_list>
#define __cpp_lib_int_pow2 202002L // freestanding, also in <bit>
[...]
```

```
#define __cpp_lib_unwrap_ref 201811L // freestanding, also in <type_traits>
#define __cpp_lib_valarray YYYYMMML // also in <valarray>
#define __cpp_lib_variant 202306L // also in <variant>
```

§ 4.2. [valarray.syn]

Modify [valarray.syn] as follows:

[...]

```
template<class T> valarray<T> tan (const valarray<T>&);  
template<class T> valarray<T> tanh (const valarray<T>&);  
  
template<class T> unspecified1 begin(valarray<T>& v);  
template<class T> unspecified2 begin(const valarray<T>& v);  
template<class T> unspecified1 end(valarray<T>& v);  
template<class T> unspecified2 end(const valarray<T>& v);  
}
```

[...]

3. Any function returning a `valarray<T>` is permitted to return an object of another type, provided all the const member functions of `valarray<T>` other than `begin` and `end` are also applicable to this type. This return type shall not add more than two levels of template nesting over the most deeply nested argument type.

4. Implementations introducing such replacement types shall provide additional functions and operators as follows:

- (4.1) for every function taking a const `valarray<T>&` other than `begin` and `end`, identical functions taking the replacement types shall be added;
- (4.2) for every function taking two const `valarray<T>&` arguments, identical functions taking every combination of const `valarray<T>&` and replacement types shall be added.

5. In particular, an implementation shall allow a `valarray<T>` to be constructed from such replacement types and shall allow assignments and compound assignments of such types to `valarray<T>`, `slice_array<T>`, `gslice_array<T>`, `mask_array<T>` and `indirect_array<T>` objects.

[...]

§ 4.3. [template.valarray.overview]

NOTE: R1 proposed the `iterator` and `const_iterator` typedefs as exposition-only, but since LEWG didn't seem to object, R2 makes them non-exposition-only.

NOTE: We propose that `valarray`'s `.begin()` should be non-noexcept, for consistency with `.size()`. Adding `noexcept` consistently throughout `<valarray>` would be cool, but is out of scope.

Modify [template.valarray.overview] as follows:

```

namespace std {
    template<class T> class valarray {
    public:
        using value_type = T;
        using iterator = unspecified;
        using const_iterator = unspecified;

        // [valarray.cons], construct/destroy
        valarray();
        explicit valarray(size_t);

    [...]

        // [valarray.range], range access

        iterator begin().;
        iterator end().;
        const iterator begin() const;
        const iterator end() const;

        // [valarray.members], member functions
        void swap(valarray&) noexcept;

        size_t size() const;

        T sum() const;
        T min() const;
        T max() const;

        valarray shift (int) const;
        valarray cshift(int) const;
        valarray apply(T func(T)) const;
        valarray apply(T func(const T&)) const;
        void resize(size_t sz, T c = T());
    };

```

§ 4.4. [valarray.members]

Move the existing section [\[valarray.range\]](#) from its current location to make it a sibling of [\[valarray.members\]](#); then modify it as follows:

~~28.6.10~~ 28.6.2.x **valarray** range access [valarray.range]

1. In the `begin` and `end` function templates that follow, *unspecified1* is a type that **The iterator** type meets the requirements of a mutable `Cpp17RandomAccessIterator` ([random.access.iterators]) and models `contiguous_iterator` ([iterator.concept.contiguous]), whose `value_type` is the template parameter `T` and whose `its` reference type is `T&`. *unspecified2* is a type that **The const iterator** type meets the requirements of a constant `Cpp17RandomAccessIterator` and models `contiguous_iterator`; whose `its` `value_type` is the template parameter `T` and whose `its` reference type is `const T&`.

2. The iterators returned by `begin` and `end` for an array are guaranteed to be valid until the member function `resize(size_t, T)` is called for that array or until the lifetime of that array ends, whichever happens first.

```
template<class T> unspecified1 begin(valarray<T>& v);  
template<class T> unspecified2 begin(const valarray<T>& v);  
iterator begin();  
const_iterator begin() const;
```

3. Returns: An iterator referencing the first value in the array.

```
template<class T> unspecified1 end(valarray<T>& v);  
template<class T> unspecified2 end(const valarray<T>& v);  
iterator end();  
const_iterator end() const;
```

4. Returns: An iterator referencing one past the last value in the array.

28.6.2.8 Member functions [valarray.members]

```
void swap(valarray& v) noexcept;
```

1. Effects: `*this` obtains the value of `v`. `v` obtains the value of `*this`.

2. Complexity: Constant.

§ 4.5. [support.initlist]

Modify [support.initlist] as follows:

[...]

17.10.2 Header `<initializer_list>` synopsis [initializer.list.syn]

```
namespace std {
    template<class E> class initializer_list {
    public:
        using value_type      = E;
        using reference        = const E&;
        using const_reference = const E&;
        using size_type        = size_t;

        using iterator         = const E*;
        using const_iterator   = const E*;

        constexpr initializer_list() noexcept;

        constexpr const E* data() const noexcept;
        constexpr size_t size() const noexcept;    // number of elements
        [[nodiscard]] constexpr bool empty() const noexcept;
        constexpr const E* begin() const noexcept; // first element
        constexpr const E* end() const noexcept;   // one past the last element
    };

    // [support.initlist.range], initializer list range access
    template<class E> constexpr const E* begin(initializer_list<E> il) noexcept;
    template<class E> constexpr const E* end(initializer_list<E> il) noexcept;
}
```

1. An object of type `initializer_list<E>` provides access to an array of objects of type `const E`.

[Note: A pair of pointers or a pointer plus a length would be obvious representations for `initializer_list`. `initializer_list` is used to implement initializer lists as specified in [dcl.init.list]. Copying an `initializer_list` does not copy the underlying elements. — end note]

2. If an explicit specialization or partial specialization of `initializer_list` is declared, the program is ill-formed.

17.10.3 Initializer list constructors [support.initlist.cons]

```
constexpr initializer_list() noexcept;
```

1. Postconditions: `size() == 0`.

17.10.4 Initializer list access [support.initlist.access]

```
constexpr const E* begin() const noexcept;
```

1. Returns: A pointer to the beginning of the array. If `size() == 0` the values of `begin()` and `end()` are unspecified but they shall be identical.

```
constexpr const E* end() const noexcept;
```

2. Returns: `begin() + size()`.

```
constexpr const E* data() const noexcept;
```

x. Returns: `begin()`.

```
constexpr size_t size() const noexcept;
```

3. Returns: The number of elements in the array.

4. Complexity: Constant ~~time~~.

```
[[nodiscard]] constexpr bool empty() const noexcept;
```

x. Returns: `size() == 0`.

~~17.10.5 Initializer list range access [support:inithist.range]~~

~~template<class E> constexpr const E* begin(initializer_list<E> il) noexcept;~~

~~1. Returns: `il.begin()`.~~

~~template<class E> constexpr const E* end(initializer_list<E> il) noexcept;~~

~~2. Returns: `il.end()`.~~

§ 4.6. [iterator.synopsis]

Modify [\[iterator.synopsis\]](#) as follows:

25.2 Header synopsis [iterator.synopsis]

```
#include <compare>           // see [compare.syn]
#include <concepts>          // see [concepts.syn]
#include <initializer_list>   // see [initializer.list.syn]

namespace std {

[...

    // [iterator.range], range access
    template<class C> constexpr auto begin(C& c) -> decltype(c.begin());
    template<class C> constexpr auto begin(const C& c) -> decltype(c.begin());
    template<class C> constexpr auto end(C& c) -> decltype(c.end());
    template<class C> constexpr auto end(const C& c) -> decltype(c.end());
    template<class T, size_t N> constexpr T* begin(T (&array)[N]) noexcept;
    template<class T, size_t N> constexpr T* end(T (&array)[N]) noexcept;
    template<class C> constexpr auto cbegin(const C& c)
        noexcept(noexcept(std::begin(c))) -> decltype(std::begin(c));
    template<class C> constexpr auto cend(const C& c)
        noexcept(noexcept(std::end(c))) -> decltype(std::end(c));
    template<class C> constexpr auto rbegin(C& c) -> decltype(c.rbegin());
    template<class C> constexpr auto rbegin(const C& c) -> decltype(c.rbegin());
    template<class C> constexpr auto rend(C& c) -> decltype(c.rend());
    template<class C> constexpr auto rend(const C& c) -> decltype(c.rend());
    template<class T, size_t N> constexpr reverse_iterator<T*> rbegin(T (&array)[N])
    template<class T, size_t N> constexpr reverse_iterator<T*> rend(T (&array)[N]);
    template<class E> constexpr reverse_iterator<const E*>
        rbegin(initializer_list<E> il);
    template<class E> constexpr reverse_iterator<const E*>
        rend(initializer_list<E> il);
    template<class C> constexpr auto
        crbegin(const C& c) -> decltype(std::rbegin(c));
    template<class C> constexpr auto
        crend(const C& c) -> decltype(std::rend(c));

    template<class C> constexpr auto
        size(const C& c) -> decltype(c.size());
    template<class T, size_t N> constexpr size_t
        size(const T (&array)[N]) noexcept;

    template<class C> constexpr auto
        ssize(const C& c)
            -> common_type_t<ptrdiff_t, make_signed_t<decltype(c.size())>>;
    template<class T, ptrdiff_t N> constexpr ptrdiff_t
        ssize(const T (&array)[N]) noexcept;

    template<class C> [[nodiscard]] constexpr auto
        empty(const C& c) -> decltype(c.empty());
    template<class T, size_t N> [[nodiscard]] constexpr bool
        empty(const T (&array)[N]) noexcept;
    template<class E> [[nodiscard]] constexpr bool
        empty(initializer_list<E> il) noexcept;

    template<class C> constexpr auto data(C& c) -> decltype(c.data());
    template<class C> constexpr auto data(const C& c) -> decltype(c.data());
    template<class T, size_t N> constexpr T* data(T (&array)[N]) noexcept;
    template<class E> constexpr const E* data(initializer_list<E> il) noexcept;
}
```

§ 4.7. [iterator.range]

Modify [\[iterator.range\]](#) as follows:

1. In addition to being available via inclusion of the `<iterator>` header, the function templates in `[iterator.range]` are available when any of the following headers are included: `<array>`, `<deque>`, `<flat_map>`, `<flat_set>`, `<forward_list>`, `<list>`, `<map>`, `<regex>`, `<set>`, `<span>`, `<string>`, `<string_view>`, `<unordered_map>`, `<unordered_set>`, `<valarray>`, and `<vector>`.

[...]

```
template<class E> [[nodiscard]] constexpr bool empty(initializer_list<E> il) noexcept;
```

~~22. Returns: `il.size()` == 0.~~

[...]

```
template<class E> constexpr const E* data(initializer_list<E> il) noexcept;
```

~~25. Returns: `il.begin()`.~~

§ 4.8. .data+.size cleanup

§ 4.8.1. [string.cons]

Modify [\[string.cons\]](#) as follows:

```
constexpr basic_string& operator=(initializer_list<charT> il);
```

36. *Effects:* Equivalent to:

```
return *this = basic_string_view<charT, traits>(il.begin() il.data(), il.size());
```

§ 4.8.2. [string.append]

Modify [\[string.append\]](#) as follows:

```
constexpr basic_string& append(initializer_list<charT> il);
```

16. *Effects:* Equivalent to: `return append(il.begin() il.data(), il.size());`

§ 4.8.3. [string.assign]

Modify [\[string.assign\]](#) as follows:

```
constexpr basic_string& assign(initializer_list<charT> il);
```

12. *Effects:* Equivalent to: `return assign(il.begin() il.data(), il.size());`

§ 4.8.4. [string.replace]

Modify [string.replace] as follows:

```
constexpr basic_string& replace(const_iterator i1, const_iterator i2, initializer_list<charT> il);
```

12. *Effects*: Equivalent to: `return replace(i1, i2, il.begin() il.data(), il.size());`

§ 4.8.5. [span.cons]

Modify [span.cons] as follows:

```
constexpr explicit(extent != dynamic_extent) span(std::initializer_list il);
```

18. *Constraints*: `is_const_v<element_type>` is true.

19. *Preconditions*: If `extent` is not equal to `dynamic_extent`, then `il.size()` is equal to `extent`.

20. *Effects*: Initializes `data_` with ~~`il.begin()`~~ `il.data()` and `size_` with `il.size()`.

§ 4.8.6. [valarray.cons]

Modify [valarray.cons] as follows:

```
valarray(initializer_list<T> il);
```

9. *Effects*: Equivalent to `valarray(il.begin() il.data(), il.size())`.

§ 4.9. Resolutions for LWG3624 and LWG3625

Modify [iterator.range] as follows:

1. In addition to being available via inclusion of the `<iterator>` header, the function templates in [iterator.range] are available when any of the following headers are included: `<array>`, `<deque>`, `<flat_map>`, `<flat_set>`, `<forward_list>`, `<list>`, `<map>`, `<regex>`, `<set>`, `<span>`, `<stacktrace>`, `<string>`, `<string_view>`, `<unordered_map>`, `<unordered_set>`, `<valarray>`, and `<vector>`.

Modify [iterator.synopsis] as follows:

```
#include <compare>           // see [compare.syn]
#include <concepts>          // see [concepts.syn]
#include <initializer_list> // see [initializer.list.syn]
```

Modify [any.synop] as follows (since [any.class] requires both `std::initializer_list` and `std::type_info`):

```
#include <initializer_list> // see [initializer.list.syn]
#include <typeinfo> // see [typeinfo.syn]

namespace std {
```

Modify [\[functional.syn\]](#) as follows (since [\[func.wrap.move\]](#) requires `std::initializer_list` and [\[func.wrap.func\]](#) requires `std::type_info`):

```
#include <initializer_list> // see [initializer.list.syn]
#include <typeinfo> // see [typeinfo.syn]

namespace std {
```

Contra LWG3624, do not modify [\[type.index.synopsis\]](#); its only use of `std::type_info` is exposition-only.

Contra LWG3624, do not modify [\[stacktrace.syn\]](#). It uses `std::initializer_list` indirectly, via [\[iterator.range\]/1](#) — but it uses `std::reverse_iterator` indirectly in the same way, and we certainly don't expect it to include `<iterator>`! The exact mechanism by which the library vendor satisfies [\[dcl.init.list\]/2](#) should be left unspecified in this case.

§ References

§ Informative References

[CWG2825]

Arthur O'Dwyer. *Range-based for statement using a braced-init-list*. November 2023. URL: <https://cplusplus.github.io/CWG/issues/2825.html>

[LWG3624]

Jiang An. *Inconsistency of <typeinfo>, <initializer_list>, and <compare>*. October 2021. URL: <https://cplusplus.github.io/LWG/issue3624>

[LWG3625]

Jiang An. *Should <stacktrace> provide range access function templates?*. October 2021. URL: <https://cplusplus.github.io/LWG/issue3625>

[LWG4001]

Hewill Kang. *iota_view should provide empty*. October 2023. URL: <https://cplusplus.github.io/LWG/issue4001>

[LWG4035]

Hewill Kang. *single_view should provide empty*. December 2023. URL: <https://cplusplus.github.io/LWG/issue4035>

[P2613]

Yihe Li. *Add the missing empty to mdspan*. June 2022. URL: <https://www.open-std.org/jtc1/sc22/wg21/docs/papers/2022/p2613r1.html>